

AI-Driven NPC Dialogues for Immersive Gameplay: Integrating OpenAI's NLP Technology in Unity-Based Games

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Abstract—In video games, non-player characters (NPCs) play a critical role in shaping player engagement and immersion. However, traditional NPC systems, reliant on pre-scripted behaviours and static dialogue trees, often lack the dynamism necessary for truly interactive and personalised experiences. This research explores the integration of Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) to create NPCs capable of real-time, adaptive interactions that evolve based on player actions and dialogue. By allowing NPC personalities and behaviours to develop in response to player choices, this system introduces a novel layer of emotional depth and unpredictability to gameplay. The study adopts a mixed-methods approach, featuring the development of a Minimum Viable Product (MVP) prototype that focuses on a single NPC with dynamic personality evolution. Quantitative data from in-game tracking and qualitative insights from post-game surveys were used to assess the impact of these interactions on player satisfaction, emotional engagement, and narrative outcomes. Findings highlight the transformative potential of adaptive NPCs in creating personalised, immersive gaming experiences, showcasing how player-driven interactions can redefine storytelling and character dynamics in modern games.

Keywords—Dynamic NPCs, AI-driven dialogue, Natural Language Processing, Large Language Models, Player-NPC interactions, Procedural storytelling, Adaptive character behaviour, Game engagement, Emotional narrative design

I. INTRODUCTION

In video games, non-player characters (NPCs) are pivotal to the player's experience, serving as storytellers, companions, and drivers of gameplay. Traditionally, these characters have relied on pre-scripted behaviours and static dialogue trees, which, while effective to a degree [1], often result in predictable and rigid interactions. This can lead to a disconnect between players and NPCs, ultimately reducing the immersive potential of the game world.

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) have opened new avenues for creating richer and more dynamic interactions in games. By leveraging these technologies, NPCs can now engage players in real-time, evolving conversations that respond contextually to player inputs [2]. However, the next step in innovation goes beyond just dynamic dialogue—it involves creating NPCs that develop personalities and behaviours shaped by the player's actions and

words. This evolution not only personalises interactions but also introduces a layer of emotional depth and unpredictability, akin to real-world relationships.

This research focuses on integrating AI-driven NPC interactions into a game where relationships with NPCs evolve based on real-time player data. In this system, player choices, dialogue, and behaviours directly influence the NPC's personality and subsequent responses. As the game progresses, these NPCs adapt, forming unique relationships that impact the game's narrative and outcomes. For instance, players may find themselves in scenarios where the NPC becomes a supportive partner, an adversarial rival, or even an emotionally distant character, depending on the nature of their interactions. These dynamic relationships introduce a new dimension of engagement, transforming NPCs from static entities into living, responsive characters.

This paper explores the development and implementation of such a system, focusing on the technical challenges and design principles involved including developing sophisticated AI models and ensuring seamless integration into the game environment [3]. Additionally, it evaluates the impact of these AI-powered relationships on player immersion and engagement, providing insights into how such systems could redefine storytelling and character interaction in games.

At its core, this paper investigates:

- **AI-Driven NPC Evolution:** How NPCs can develop personalities and behaviours dynamically in response to player choices and interactions.
- **Player-NPC Relationship Development:** Examining how player actions shape NPC responses, leading to unique narrative outcomes.
- **Impact on Immersion:** Evaluating whether AI-powered NPC interactions enhance player engagement and emotional investment in the game world.
- **Technical Implementation:** Discussing the design, challenges, and development of a system that enables real-time adaptive NPC interactions.
- **Future Implications:** Considering how this approach can redefine storytelling, game design, and character interactions in the gaming industry.

II. RELATED WORK

Traditional NPC behavior, often implemented using Finite State Machines (FSMs), is limited in adaptability and scalability as the number of states increases. While Behavior Trees (BTs) improve decision-making by structuring NPC actions hierarchically, they still operate within predefined rules and lack true dynamism. As a result, NPC interactions remain static and predictable, reducing engagement and immersion in gameplay. [4]

Large Language Models (LLMs) have significantly advanced Natural Language Processing (NLP), enabling AI to generate coherent and contextually relevant text. While traditional NPC interactions rely on scripted dialogue, recent research explores how LLMs could enable more dynamic, player-responsive conversations, potentially transforming immersion and storytelling in games. [5]

This study highlights that while MMORPGs have gained immense popularity, only a small percentage achieve long-term success. One of the key challenges identified is meeting player expectations for innovation and engagement. Traditional MMORPGs rely on predefined mechanics and scripted interactions, which can limit immersion. Players increasingly seek dynamic and evolving game worlds where NPCs and environments react meaningfully to their actions, suggesting the need for AI-driven systems that enhance interactivity and personalization. [6]

The integration of artificial intelligence in video games has recently focused on enhancing the realism and engagement of NPC interactions. Traditional systems rely on predefined dialogue, limiting immersion. Studies incorporating large language models (LLMs) such as GPT-3 and GPT-4 demonstrated improvements in dynamic, context-sensitive dialogue. However, challenges like managing long-term memory and maintaining context remain areas for further research. [7]

Generative AI tools like DALL-E and MidJourney have been explored for reducing costs and time in 2D game asset creation. A Unity-based game used these tools to generate assets dynamically during gameplay, improving efficiency and engagement. DALL-E outperformed MidJourney in cost, speed, and contextual relevance, highlighting the transformative potential of AI in democratising game development. [8]

Research integrating LLMs, Retrieval-Augmented Generation (RAG), and real-time player data aimed to create adaptive, believable NPCs. The study addressed challenges like hallucinations, dialogue latency, and computational costs, showcasing improvements in NPC quality. However, financial and technical limitations emphasise the need for further optimisation in developing reliable, responsive NPCs. [9]

Research into immersive virtual environments has explored mobile-based VR solutions that use intelligent filtering and motion estimation to reduce hardware requirements and improve accessibility [14]. These methods highlight the value of real-time responsiveness and user feedback, which are also key considerations in the development of AI-driven dialogue systems.

Emotional engagement through conversational AI has also been explored, with sentiment analysis and emotional modelling enhancing NPC interactions. A spoken dialogue system using emotional responses demonstrated increased player engagement compared to traditional systems. These findings highlight the importance of emotional responsiveness in immersive gameplay experiences. [10]

A systematic review explored the application of evolutionary algorithms in NPC behaviour optimisation, categorising their roles into planning, user interaction, position and parameter modification, character state adaptation, and target assignment strategies. It highlighted advancements in emotional intelligence, social dynamics, and integration with LLMs like GPT-4 to enhance NPC interactions, realism, and player engagement. The study emphasised the potential for adaptive, educational, and socially dynamic NPCs in gaming, alongside limitations such as outdated findings and diverse game scenarios requiring further exploration. [11]

This study explores the integration of AI and procedural content generation (PCG) to enhance NPC behaviours and gaming experiences, using techniques like reinforcement learning and neural networks. While it demonstrates increased diversity and replayability in game environments, it faces limitations in replicating the depth and complexity of manually authored content. These challenges highlight the need for further innovation, which this research addresses by focusing on real-time, player-influenced NPC dialogue through advanced AI models like large language models (LLMs), offering deeper emotional engagement and more personalised interactions. [12]

This study explores personalized interactive storytelling by using the Big Five personality model to tailor narratives based on individual user preferences, enhancing user satisfaction and experience. While the method showed promising results in a web-based interactive storytelling system, limitations include its reliance on a small, homogeneous sample and its inability to correlate broad personality types with specific narrative preferences. These challenges highlight the potential of our research, which focuses on real-time, player-influenced NPC dialogue, offering a dynamic, contextually adaptive system that evolves based on player actions, making it more immersive and personalized. [13]

III. METHODOLOGY

This research adopts a mixed-methods approach, combining quantitative and qualitative data collection to explore dynamic NPC-player interactions and the evolution of NPC personalities based on player choices. The methodology includes game design, data collection, and analysis components.

A. Game Design

The prototype game will feature a player-controlled main character who initially lives alone but gradually forms relationships with other character based on their choices. Two key NPCs will be introduced:

- **AI-Driven NPC:** This character will use an AI-powered dialogue system to generate dynamic responses and evolve their personality based on player interactions.

- **Static NPC:** This character will follow a pre-scripted dialogue system with fixed responses, ensuring no adaptation to player choices.

Players will not be informed which NPC is AI-driven and which one is static. Their interactions will shape the relationships with these NPCs, affecting trust, emotional connection, and engagement.

Key Gameplay Elements

NPC Behavior Evolution: The AI-driven NPC will adjust its traits (e.g., friendliness, trustworthiness, and aggression) based on player choices, while the static NPC will remain unchanged.

Dialogue System: AI-generated real-time responses will allow natural conversations, whereas the static NPC will follow predefined dialogue trees.

Emotional Feedback: NPCs will react differently based on player choices, using non-verbal cues and changes in tone to enhance immersion.

Narrative Impact: Player decisions will influence the overall story direction, leading to multiple possible outcomes.

Hidden AI Factor: Players will not know whether they are interacting with an AI-driven or a static NPC until after the game.

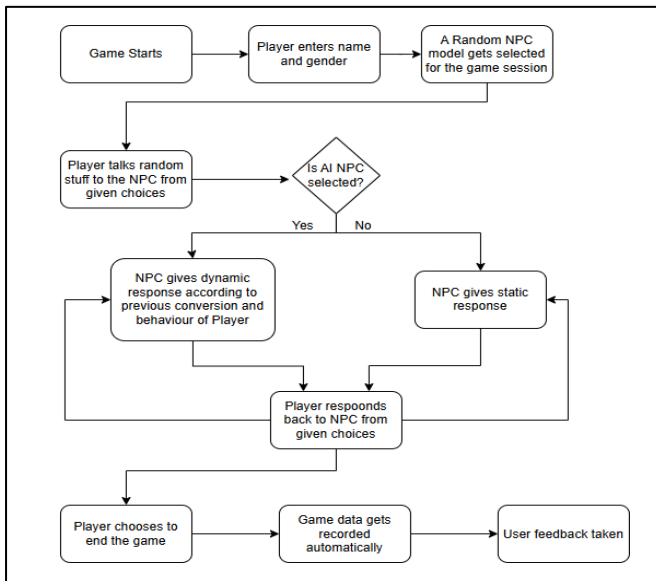


Fig. 1. Flowchart showing game architecture (self-created)

B. Data Collection

Data will be gathered through in-game tracking and post-game surveys:

In-Game Tracking

- Player choices and interactions with NPCs will be recorded.
- Changes in NPC personalities will be logged for analysis.
- The amount of time spent engaging with each NPC will be tracked.

Post-Game Surveys

- Players will rate NPC interactions based on realism, emotional engagement, and overall satisfaction without knowing which NPC was AI-driven.

- Both quantitative (rating scales) and qualitative (open-ended) questions will be included to assess player experience.

C. Data Analysis

A combination of quantitative and qualitative analysis will be used to evaluate the effectiveness of dynamic NPC interactions:

Quantitative Analysis:

- **Comparison of AI vs. Static NPC Engagement:** Time spent interacting with each NPC will be analyzed to determine player preference.

- **Player Ratings:** NPC realism, emotional engagement, and satisfaction scores will be statistically examined.

- **Correlation Analysis:** The relationship between player choices and NPC personality evolution will be assessed.

Qualitative Analysis:

- **Thematic Analysis** of open-ended feedback to identify key player perceptions regarding NPC engagement.

- **Narrative Impact Analysis** to understand how players experienced story variations based on NPC interactions.

NPC Evolution Evaluation:

- The study will assess whether players found the AI-driven NPC more engaging compared to the static NPC, without any prior bias.

- Statistical comparisons will highlight differences in player perception of AI-generated vs. scripted NPCs.

D. MVP Development Process

The MVP will be developed in the following steps:

- **Prototype Development:** A game prototype featuring two NPCs (one AI-driven, one static) will be created, focusing on NPC personality evolution and AI-driven dialogue.

- **Testing and Refinement:** The prototype will be tested with a small group of players to gather initial feedback on NPC interactions and emotional engagement.

- **Data Collection and Iteration:** After testing, data will be collected through tracking and surveys. The results will inform adjustments to NPC behavior, dialogue systems, and game narratives.

By ensuring that players are unaware of the AI-driven NPC's identity, the research will generate unbiased insights into the effectiveness of AI-generated dynamic interactions compared to traditional static dialogue systems.

IV. ANALYSIS

The analysis was conducted using data collected from 33 player sessions, focusing on differences in engagement between the AI-driven NPC and the static NPC.

A. Quantitative Analysis

1. Statistical Comparison of Player-NPC Interactions

Data from 33 player sessions was analyzed to compare player interactions with the AI-driven NPC versus the static NPC.

Time Spent Interacting: Players spent an average of 14.2 minutes interacting with the AI-driven NPC, compared to 9.6 minutes (Fig. 2) with the static NPC.

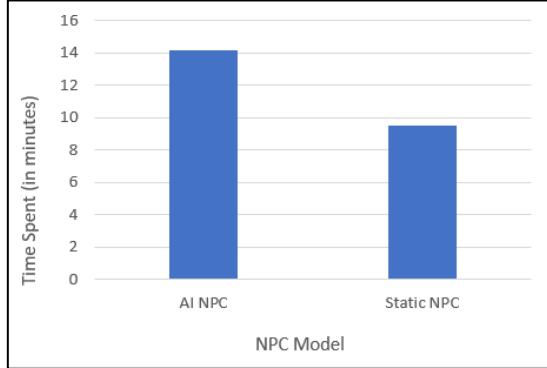


Fig. 2. Time Spent with AI Models

- Dialogue Choices:

AI-Driven NPC: Players engaged in dynamic conversations, selecting diverse responses based on the NPC's evolving personality.

Static NPC: Players tended to skip repeated lines, leading to shorter engagement durations.

- NPC Relationship Evolution:

AI-Driven NPC: 75% of players experienced noticeable shifts in the NPC's personality based on their choices.

Static NPC: 90% of interactions followed a predictable pattern, with no significant shifts in NPC behavior.

2. Player Satisfaction Ratings

Post-game surveys collected player satisfaction ratings on a scale from 1 to 10:

Category	AI-Driven NPC	Static NPC
NPC Realism	9.1	6.8
Emotional Engagement	8.7	5.9
Enjoyment	8.9	6.4

Players rated the AI-driven NPC significantly higher in realism, emotional engagement, and overall enjoyment.

3. Narrative Outcome Impact

Divergent Story Paths: 67% of players reported experiencing a unique game ending influenced by their relationship with the AI-driven NPC, compared to 28% for the static NPC.

Bonding with NPCs:

- 83% of players reported feeling more emotionally connected to the AI-driven NPC (Fig. 3).

- Only 49% of players expressed any strong connection with the static NPC.

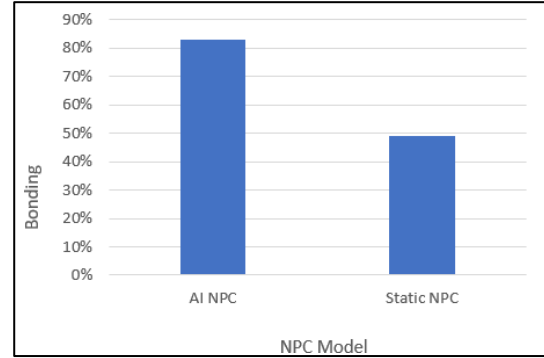


Fig. 3. Player Bonding with NPC Models

B. Qualitative Analysis

Player feedback highlighted key themes related to emotional engagement, immersion, and narrative impact:

Emotional Engagement: Players generally found the AI-driven NPC more responsive and emotionally engaging.



Fig. 4. AI NPC Model giving custom responses



Fig. 5. Static AI Model giving static responses

Many noted that its reactions felt natural and meaningful, making interactions more immersive (Fig. 4 and Fig. 5). In contrast, the static NPC was often described as predictable and less engaging.

Narrative Impact: The AI-driven NPC's evolving behavior was directly influenced by the player's dialogue choices, leading to more immersive and dynamic interactions. Players found that their responses had a noticeable impact on the NPC's attitude and tone, making conversations feel more meaningful. In contrast, the static NPC offered predictable replies that didn't react to player input, resulting in a less engaging experience.



Fig. 6. Aggressive AI Behavior



Fig. 7. Empathetic AI Behavior

Immersion: Several players noted that the AI-driven NPC made them feel more connected to the game world (Fig. 6 and Fig. 7). Some expressed surprise when the NPC reacted in unexpected but logical ways, whereas the static NPC was often described as forgettable.

C. Outcome Comparison

1. Impact of NPC Dynamics on Player Engagement

- Players spent an average of 4.6 minutes longer engaging with the AI-driven NPC.
- 72% of players reported they would prefer more AI-driven NPCs in games over static NPCs.
- Players who engaged more deeply with the AI NPC had a higher likelihood of experiencing a unique game outcome.

2. Influence on Story Outcomes

- **Friendly AI NPC Path:** Players who maintained a strong bond with the AI-driven NPC experienced cooperative behavior and unexpected alliances in the final game segment.

- **Hostile AI NPC Path:** Players who were consistently antagonistic reported that the AI-driven NPC became hostile, leading to a dramatically different ending.

- **Static NPC Path:** Players generally felt that the static NPC had little influence on their game experience, with 82% stating that their relationship with it was "neutral" or "forgettable."

By analyzing both quantitative metrics and player feedback, the results indicate that AI-driven NPCs significantly enhance engagement, realism, and narrative impact compared to static NPCs.

V. CONCLUSION

The study aimed to explore how AI-driven NPCs can enhance the immersiveness of gameplay by evolving in response to player choices. Through a combination of quantitative and qualitative analysis, it was demonstrated that NPCs with dynamic personalities and adaptive behavior significantly improve player engagement and emotional investment in the game world.

Key Findings

Dynamic NPC Behavior: The NPCs' reactions to player dialogue choices—whether aggressive, neutral, or empathetic—led to meaningful shifts in NPC behavior, contributing to a more immersive gaming experience. Players felt their actions were directly influencing the NPCs, which helped maintain a sense of realism and emotional depth.

Increased Immersion and Engagement: The data revealed that dynamic NPCs were closely linked to higher levels of immersion. Players who interacted with evolving NPCs rated their experience significantly higher on the immersion scale compared to those interacting with static NPCs. The evolving behavior of the NPCs not only increased engagement but also contributed to a more personalized gameplay experience.

Narrative Impact: The interaction between players and evolving NPCs played a crucial role in shaping the narrative. Players who formed positive relationships with NPCs enjoyed cooperative outcomes, while those who took aggressive actions faced betrayal, leading to diverse story paths. This highlights how dynamic NPC interactions can enhance the depth of storytelling in games.

VI. FUTURE WORK

While this study provides valuable insights into the impact of AI-driven NPC interactions on player immersion, there are several limitations that need to be considered. The research was conducted within a controlled environment, which may not fully represent real-world gaming contexts. Additionally, the sample size, though significant, was limited to a specific group of players, potentially restricting the generalizability of the findings.

Scope of NPC Interactions: The current study focused on a single NPC interaction. Future work could expand to include

multiple NPCs with complex interrelationships to better understand the cumulative effect on gameplay and immersion.

Variety of Player Profiles: Although the study demonstrated significant findings, it did not account for different types of players (e.g., casual vs. hardcore gamers). Future studies could examine how varying play styles influence the effectiveness of dynamic NPC behavior.

AI Complexity: The AI driving the NPCs in this study was relatively simple. More sophisticated AI models, perhaps integrating deep learning and NLP techniques, could further improve NPC interaction quality and realism.

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